

A Wall-Following Method for Autonomous Driving

6.4200: Team 5: Lab 3

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Our project addresses real-world problems: how do we balance safety and efficiency when engineering AVs?

AP Associated Press News · 1d

Bill could bring Virginia's first self-driving cars and trucks by 2028

Even if self-driving vehicles become legally possible in Virginia, they'd have to get through several more rounds of ...



USA TODAY · 9d

What is Waymo? Self-driving taxis coming to this Florida city

Waymo's self-driving cars are now in Orlando, offering rides to select users. But are these autonomous vehicles safe?

HC Houston Chronicle on MSN · 10d

Waymo self-driving cars debut for select riders in Houston, two other Texas cities

The cars patrolled Houston for about nine months before the program's launch.



LL Land Line · 6d

Autonomous vehicles under fire for safety concerns, lack of transparency

If autonomous vehicles are so much safer, then why aren't the companies more...

We created a safety controller and a wall follower for a real-world autonomous vehicle.

Our safety controller stops the vehicle from colliding with obstacles.

Our wall follower maintains a constant distance from any wall without visible oscillations, handling turns and gaps.

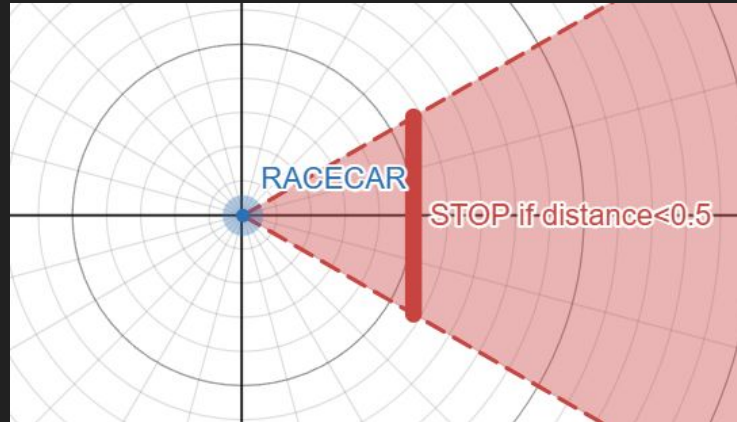


Our software works for both our simulated and physical racecar.

Our simulated safety controller is straightforward and effective.

Consider the cone from $-\pi/12$ to $+\pi/12$ to determine whether a front wall exists.

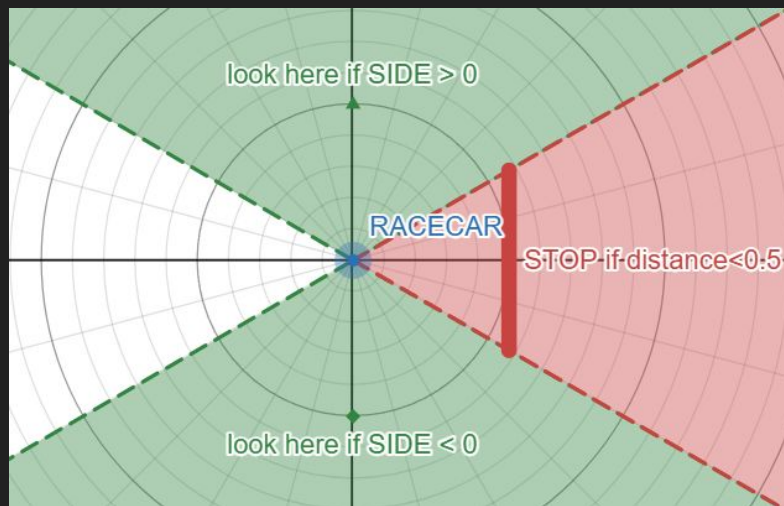
Stop if the front wall is less than 0.5 meters away.



Our approach to filtering LiDAR data takes advantage of given parameters.

Look perpendicular towards the given side of the car.

Consider the cone from $-\pi/3$ to $+\pi/3$: relatively wide to give enough data for RANSAC.

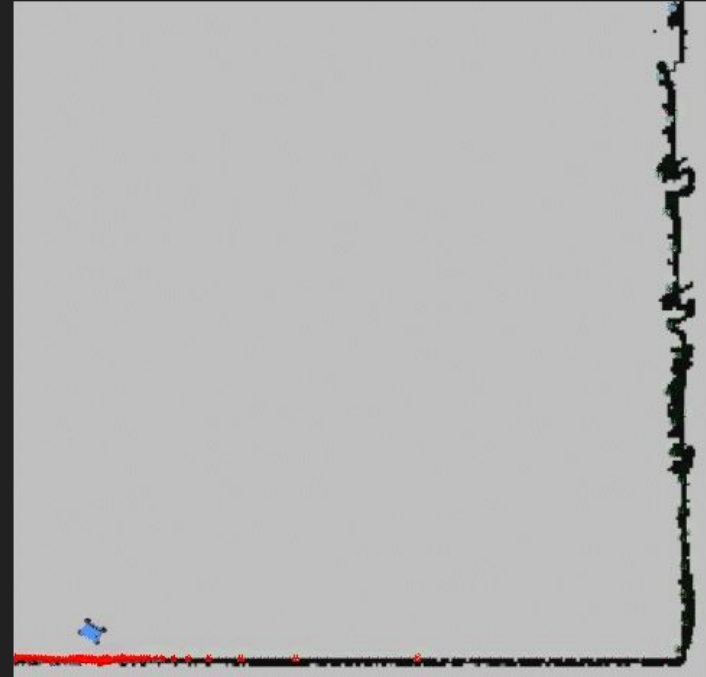


Our approach to wall modelling is accurate and efficient.

Remove outliers with 50 iterations of RANSAC (runs in linear time).

Fit a best line to the data.

Visualize!

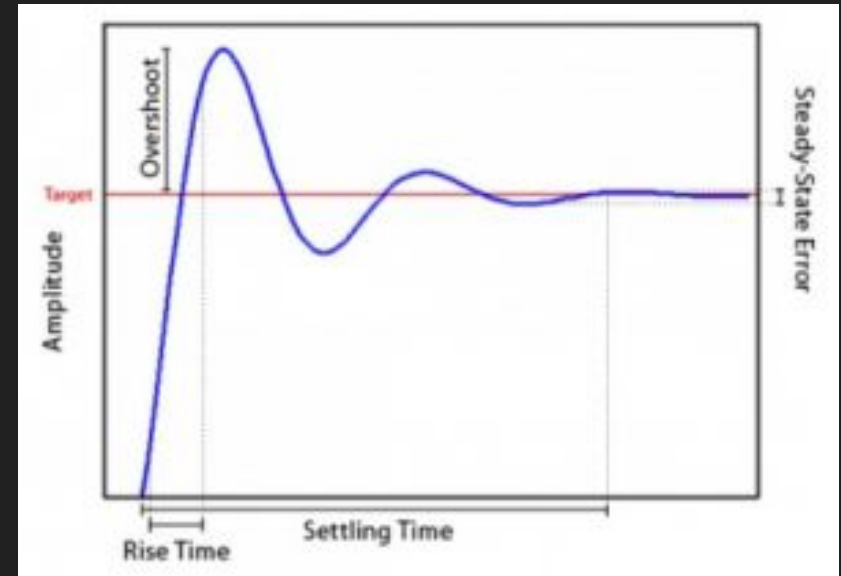


We used a proportional-derivative controller to drive our vehicle.

Straightforward Proportional-Derivative Controller to quickly adjust distance to wall while minimizing oscillations.

Constants: $K_P = 1/\text{velocity}$, $K_D = 0.5$.

Maximize steering angle at 0.34.



We took our system from simulation to the real world.

Translating to multiple levels of priority (muxes).

Trial-and-error adjustment of PD constants.

Addressing real-world edge cases: rounding external corners, surfaces with different frictions, sudden appearance of moving obstacles.



We collected a variety of relevant data to evaluate our performance.

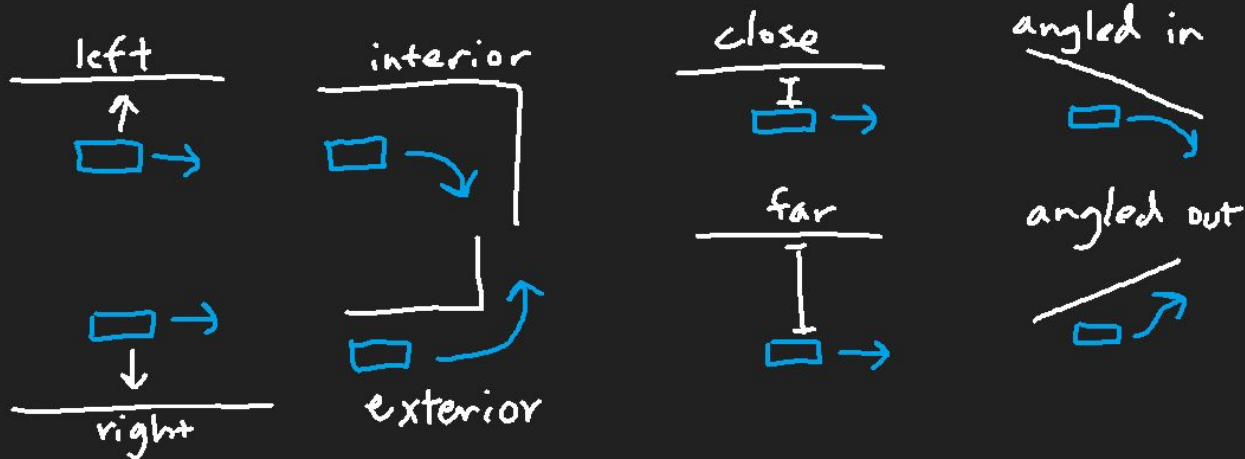
Tests:

- Left and right walls
- Interior and exterior corners
- Varying initial distance to wall
- Varying initial angle relative to wall
- Sudden obstacles

Data (average + deviation):

- Distance from wall
- Velocity
- Recorded image of trajectory

Both simulated & real-life performance.



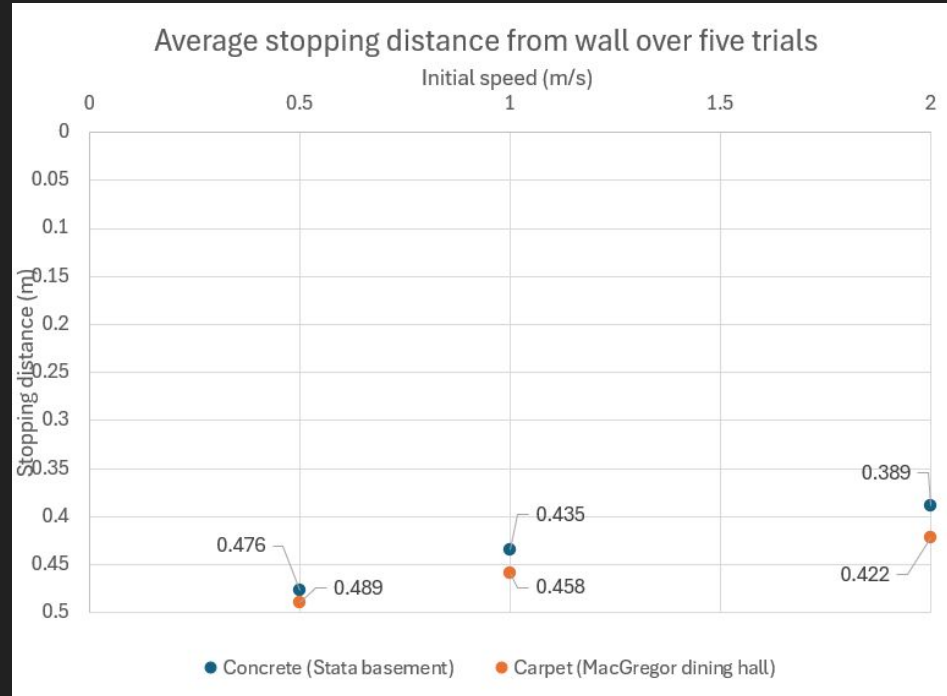
We used repeated, measurable, representative tests to evaluate our performance.

Five repetitions.

Constant starting position, angle, and surface per test.

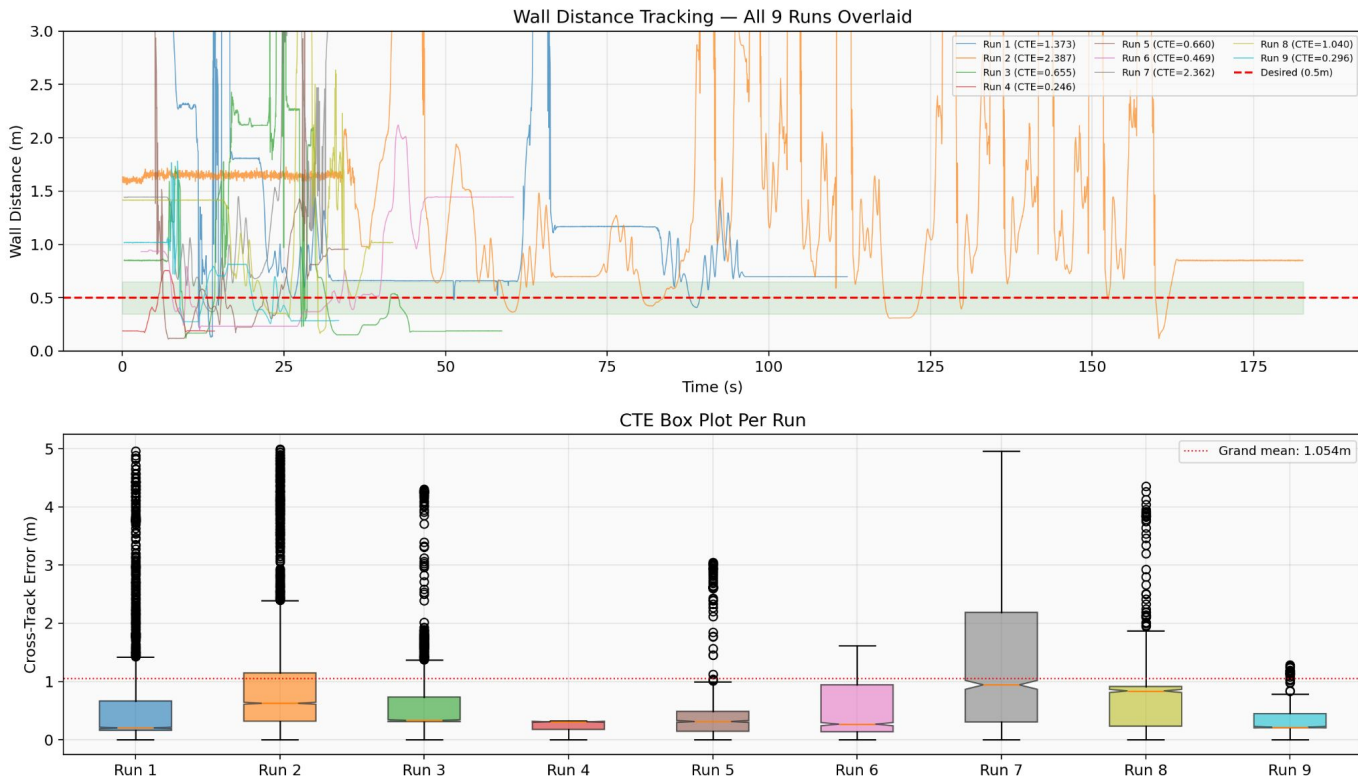
Constant random seed.

Our quantitative data shows that our autonomous vehicle controller is safe and effective.

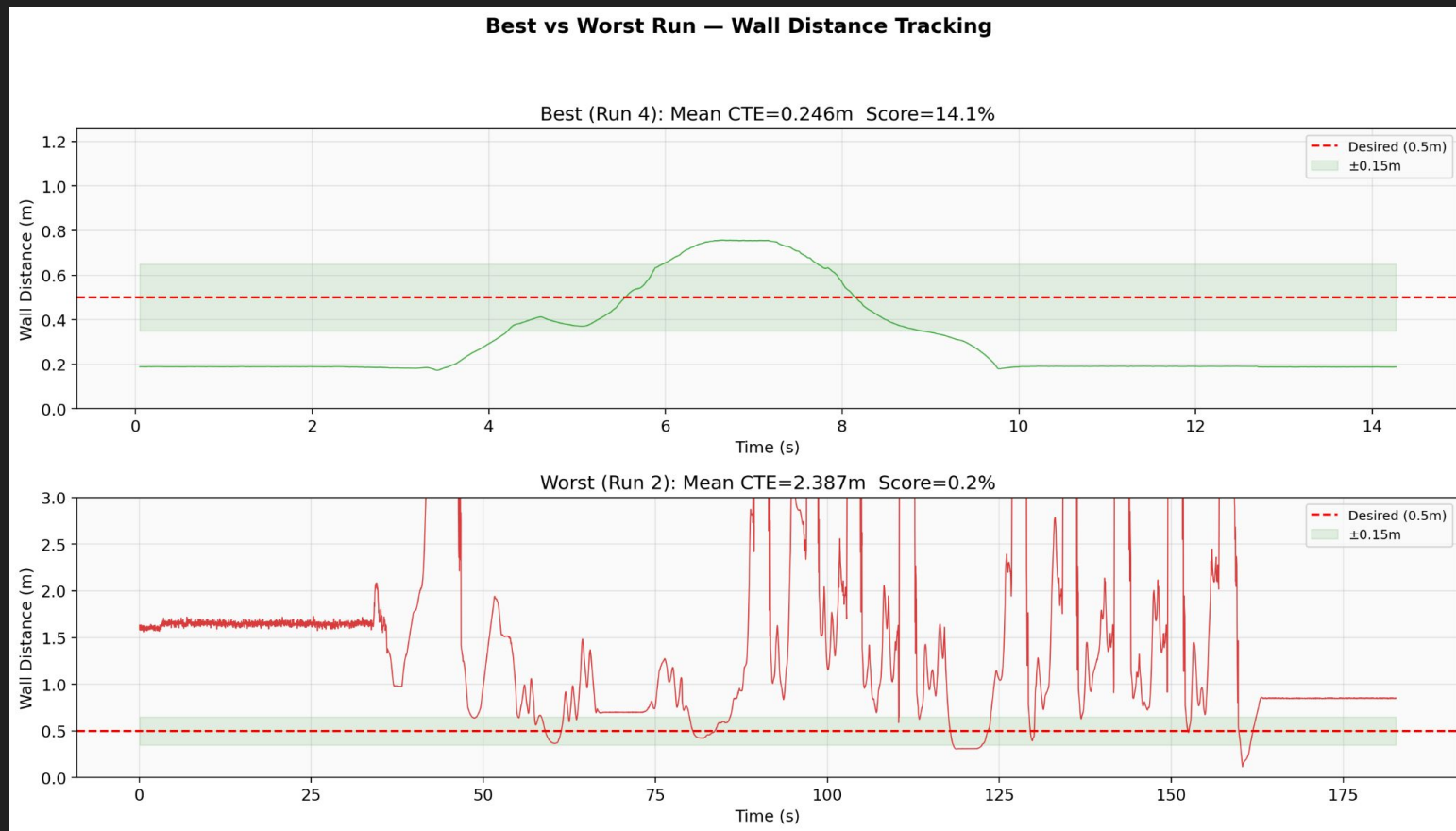


Our quantitative data shows that our autonomous vehicle controller is safe and *soon to be effective*.

Repeatability Analysis — Right Wall 0.5m (N=9 runs)



Our Best Recorded Run Reached Stable Conditions...



Our qualitative data shows that our autonomous vehicle controller is practical and safe.

No visible oscillations.

Smooth turns.

Comfortable stopping distance.



Complicated wall



Sudden moving obstacle

We still see room for further development.

More thorough evaluation of different conditions:

Uphill? Downhill?

Darkness?

Wind?

Low obstacles? Messy terrain?



Tesla Model 3 Driven into Floodwater While on Full **Self Driving**

YouTube · Vehiclesuggest
1K views · Aug 27, 2023



AI Delivery Car Goes Anywhere... Even Flooded Roads



YouTube · Mistmap
20.9K views · 1 month ago

Thank you!

